Safety Erasure Under KV-Cache Compression

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Abstract

Inference systems routinely compress or evict the key-value (KV) cache to reduce serving cost. We show that on dense full-attention instruction-tuned language models, this throughput optimization can become a selective safety intervention: refusal and policy-following behavior degrades meaningfully more than ordinary task capability when the cache is aggressively evicted. We test this across nine open-weight checkpoints from seven model families and find positive Selective Safety Erasure Index (SSEI) in four families (Llama, OLMo, Phi, Qwen; five instruction-tuned models, with effect sizes from 0.02 to 0.09 in absolute pass-rate units, 95% bootstrap CIs excluding zero) under sliding-window and user-pinned cache policies. The effect is robust to leave-one-family-out perturbation. Three models do not show the pattern, and the architectural common thread among them (interleaved or full-layer sliding-window attention; non-standard mixture-of-experts cache handling) is consistent with the hypothesis that the affected behavior depends on long-range KV cache state. Causal-patching experiments that restore baseline cache state into compressed runs show partial recovery of refusal behavior (35–58% restoration fraction, depending on model); however, both system-role and matched user-role interventions produce comparable restoration under a per-prompt mean-of-ratios estimator, so we do not find evidence that the safety-relevant information is role-specifically localized. Policy-pinned cache retention fully restores refusal.

1 Introduction

Production language-model serving systems do not deploy a static pair of weights and prompts. They actively manage the transient KV cache through eviction policies, quantization, and paged attention to fit longer contexts into a fixed memory budget (Kwon et al., 2023; Zhang et al., 2023; Xiao et al., 2024). These mechanisms are documented to degrade task quality (Chen et al., 2025; Yang et al., 2024; Ananthanarayanan et al., 2026). We ask a different question: does cache compression weaken *refusal* behavior more than it weakens ordinary capability on the same prompts and budget?

The result is yes for the majority of currently-deployed dense-attention instruction-tuned models. Across nine checkpoints sampled from Llama, Gemma, Mistral, OpenAI gpt-oss, OLMo, Phi, and Qwen families, five instruction-tuned models show a Selective Safety Erasure Index (SSEI) of at least one cache policy whose 95% lower CI exceeds zero. Phi-4 and Qwen3-9B exhibit the largest effects, around eight to nine percentage points of safety loss beyond matched capability loss under sliding-window cache eviction. The cross-family conclusion holds when any single family is removed from the panel.

Three models do not show the pattern: Gemma-2-9B-IT, Mistral-7B-Instruct-v0.3, and OpenAI gpt-oss-20b. These models share architectural features that decouple long-range KV cache state from the safety-relevant generation circuit: Gemma 2 interleaves local sliding-window attention with global attention (Gemma Team, Google DeepMind, 2024), Mistral 7B v0.3 applies sliding-window attention at every layer (Jiang et al., 2023), and gpt-oss-20b is a mixture-of-experts model trained with a structured "harmony" response format (OpenAI, 2025). Under our cache interventions gpt-oss-20b also collapses into a uniform output mode rather than gracefully degrading. Read together, the positive results in four families and the negative results in these three constitute a falsification test that the affected behavior is specifically cache-resident long-range attention state, not generic compression sensitivity.

Causal-patching experiments, in which baseline cache state is restored into compressed runs, show partial recovery of refusal behavior on Qwen2.5-7B-Instruct (single-seed) and Qwen3-9B (multi-seed, Sonnet-judged). Under a per-prompt bootstrap estimator, both system-role and matched user-role interventions produce comparable restoration fractions (Qwen3-9B: 0.355 vs. 0.408, overlapping CIs; Qwen2.5-7B: 0.584 vs. 0.584), so we do not find evidence of role-specific localization. A parallel Phi-4 run shows a null causal result: simulated four-bit quantization does not degrade Phi-4's safety in the first place (0.985 vs. 0.987 baseline), leaving insufficient denominator for meaningful restoration analysis.

Contributions. We make three contributions. First, a cross-family measurement of selective safety degradation under cache eviction across nine open-weight checkpoints, with explicit per-policy effect sizes and bootstrap confidence intervals. Second, an architecture-dependent characterization of which models are affected and which are robust, framed as a structural prediction rather than a model-level idiosyncrasy. Third, causal-patching evidence that restoring baseline cache state into compressed runs partially recovers refusal behavior, though without a role-specific dissociation between system-role and user-role tokens.

2 Related Work

KV-cache compression. Algorithms for KV-cache management trade off memory for quality. H2O retains heavy-hitter tokens (Zhang et al., 2023); StreamingLLM relies on attention sinks for long contexts (Xiao et al., 2024); SnapKV compresses prompts using observed attention patterns (Li et al., 2024); KIVI and KVQuant study low-bit quantization (Liu et al., 2024; Hooper et al., 2024). Chen et al. (2025) report that KV compression unevenly harms instruction following and increases system-prompt leakage. Ananthanarayanan et al. (2026) frame compression as a perturbation of token accessibility through attention dynamics. We extend this line of work along the safety axis: rather than measuring overall quality loss, we measure safety loss minus capability loss on matched prompts.

Cache state as a control surface. Wang et al. (2025) show that KV-cache state can be edited to defend against indirect prompt injection, and prompt-extraction work establishes that the cache can be read out as well (Hui et al., 2024). These results frame cache state as an addressable safety-relevant object. Our experiments treat cache state as a causal locus and ask whether ordinary serving-time eviction can damage that state.

Mechanism-first safety phenomena. A growing body of work isolates specific mechanisms in safety failure rather than only reporting benchmark accuracy: refusal directions (Arditi et al., 2024; Joad et al., 2026), sparse alignment routes (Frank, 2026), depth-conditioned refusal restoration (Zhang et al., 2026), subliminal learning (Cloud et al., 2025), token entanglement (Zur et al., 2025), and emergent misalignment (Betley et al., 2025). The closest is Wang et al. (2025), which uses cache state as an intervention point. We test whether cache state is a *naturally occurring* safety failure point under standard inference optimizations.

3 Method

3.1 Setup

We evaluate nine open-weight checkpoints: Qwen2.5-7B (base and Instruct), Qwen3-9B, Llama-3.1-8B-Instruct, Gemma-2-9B-IT, Mistral-7B-Instruct-v0.3, OLMo-3-7B-Instruct, Phi-4, and OpenAI gpt-oss-20b. Models are loaded in bfloat16 with explicit access to the cache state, so eviction is implemented in the generation loop rather than delegated to a serving system. We do not claim coverage of vLLM, TGI, or PagedAttention kernels; the results are intended as a cross-family measurement of the underlying behavior, and operator-level deployment validation is outside the scope of this paper.

3.2 Cache Policies

We compare uncompressed generation against five cache interventions:

- **sliding window**: retain only the most recent N cached tokens;
- **sink plus recent**: retain the first S tokens and the most recent N;
- **random matched**: evict random tokens at the same budget as a structured policy, as a non-policy control;
- **user pinned**: protect user-role spans, then fill the remaining budget with sink and recent retention. This is the non-policy control for policy pinning;
- **policy pinned**: protect system-role or policy spans, then fill the remaining budget with sink and recent retention.

For all policies, we record retained and evicted token counts by role at each decode step.

3.3 Prompt Suites

The benchmark separates safety, leakage, over-refusal, and capability:

- **Public refusal safety**: harmful requests drawn from AdvBench and JailbreakBench (Zou et al., 2023; Chao et al., 2024);
- **Public benign over-refusal**: Databricks Dolly 15k benign instructions (Databricks, 2023);
- **Public XSTest safe**: benign safety-sensitive prompts (Rottger et al., 2023);
- **Public system leakage**: prompt-injection attacks from the Cyberec corpus (Cyberec, 2026) paired with a hidden system canary;
- **Public capability ARC**: multiple-choice capability controls (Clark et al., 2018).

Each suite ships with a manifest of prompt IDs, source dataset revisions, and content hashes. Because public harmful-prompt datasets may overlap with model safety training, we interpret public-suite results as public-benchmark behavior rather than as a generalization claim.

3.4 Metrics

Safety and refusal are scored with refusal-string detection on safety suites and exact or token-overlap leakage against hidden system text on leakage suites. Capability is scored with multiple-choice accuracy on ARC-Easy. Over-refusal is the refusal rate on the benign suites.

For each cache policy p, budget b, and model m, we define the Selective Safety Erasure Index as the difference between safety degradation and capability degradation relative to the uncompressed baseline:

$$\mathrm{SSEI}(p, b, m) = \Delta_{\text{safety}}(p, b, m) - \Delta_{\text{capability}}(p, b, m). \tag{1}$$

Positive SSEI indicates that safety pass-rate declines more than capability pass-rate under the same intervention. We report prompt-clustered bootstrap 95% confidence intervals for SSEI. A model-policy condition is considered a positive instance of selective safety erasure when SSEI ≥ 0.01 and its lower CI exceeds zero.

3.5 Audit

Automated labels are imperfect. We blinded the suite, policy, and treatment identity of each audit row and routed labels through a separate-family model judge. Annotations come from Claude Sonnet 4.5 (Anthropic, 2024); family separation prevents same-family judging on any panel model. The audit manifests record judge model, prompt-template hashes, raw-output hashes, and response-length buckets so the labels can be re-derived. Across the panel, 2,595 of 2,676 attempted audit rows parsed (97%); per-model parse rates range from 91% to 100%.

3.6 Causal Restoration

For length-preserving cache perturbations such as simulated four-bit KV quantization, we run baseline (uncompressed) and compressed decoding on identical prompts and seeds, then patch selected key and value slices from the baseline cache into compressed runs at the same positions. We compare:

1. compressed decoding with no patch;
2. compressed decoding with system-role key+value slices restored from baseline;
3. compressed decoding with user-role slices restored, matched to the system-token count;
4. compressed decoding under the policy-pinned eviction policy that protects system spans throughout generation;
5. compressed decoding under the user-pinned policy as the matched non-policy control.

The user-role restoration is the central control: it shares the additional retained-token budget with the system-role patch while differing only in which role's content is restored. A causal-localization claim requires the system-role intervention to recover refusal substantially more than the matched user-role intervention.

4 Results

4.1 Cross-Family Selectivity

Figure 1 reports per-model per-policy SSEI point estimates and 95% bootstrap confidence intervals. The largest positive effects come from sliding-window cache eviction on Qwen3-9B (SSEI = 0.092, CI [0.083, 0.101]) and Phi-4 (SSEI = 0.084, CI [0.076, 0.091]). Smaller but significant positive effects appear under user-pinned eviction on Llama-3.1-8B-Instruct (0.024 [0.016, 0.031]), OLMo-3-7B-Instruct (0.026 [0.019, 0.033]), and Qwen2.5-7B-Instruct (0.017 [0.011, 0.022]).

Table 1: Cross-family selectivity (SSEI) by cache policy. Positive SSEI = safety loss exceeds capability loss. Values show point estimate ± half-width of 95% bootstrap CI.

Family	Model	sliding window	sink+recent	policy-pinned	user-pinned	random matched
Gemma	Gemma-2-9B-IT	-0.005 ± 0.003	-0.001 ± 0.002	-0.004 ± 0.002	-0.003 ± 0.005	-0.000 ± 0.003
OpenAI	GPT-OSS-20B	-0.068 ± 0.020	-0.068 ± 0.020	-0.079 ± 0.020	-0.082 ± 0.019	-0.079 ± 0.020
Llama	Llama-3.1-8B-Instruct	-0.085 ± 0.019	0.007 ± 0.005	-0.001 ± 0.006	0.024 ± 0.008	0.022 ± 0.008
Mistral	Mistral-7B-Instruct-v0.3	0.009 ± 0.008	0.001 ± 0.003	-0.004 ± 0.004	-0.008 ± 0.007	0.002 ± 0.005
OLMo	OLMo-3-7B-Instruct	-0.004 ± 0.004	-0.010 ± 0.005	-0.016 ± 0.005	0.026 ± 0.007	0.005 ± 0.005
Phi	Phi-4	0.084 ± 0.008	0.001 ± 0.002	-0.001 ± 0.002	0.055 ± 0.009	0.044 ± 0.006
Qwen	Qwen2.5-7B base	–	–	–	–	–
Qwen	Qwen2.5-7B-Instruct	0.011 ± 0.005	-0.003 ± 0.005	-0.001 ± 0.005	0.017 ± 0.006	0.012 ± 0.005
Qwen	Qwen3-9B	0.092 ± 0.009	-0.005 ± 0.004	0.010 ± 0.005	-0.002 ± 0.004	0.020 ± 0.006

The cross-family conclusion is robust to single-family removal. Table 2 reports a leave-one-family-out check: for each family, we re-evaluate the cross-family claim after excluding that family and verify that at least two of the remaining families still have a positive condition with CI excluding zero. The claim holds in every case, including when the largest-effect family (Phi) or the entire Qwen family (three checkpoints) is removed.

Table 2: Leave-one-family-out robustness check on the cross-family selectivity claim. Each row removes the named family and counts the remaining instruction-tuned families that retain positive SSEI with 95% lower CI excluding 0 on at least one registered cache policy. The registered rule requires at least two such families.

Excluded family	Positive families remaining	Claim holds?
Gemma	Llama, OLMo, Phi, Qwen	supported
Llama	OLMo, Phi, Qwen	supported
Mistral	Llama, OLMo, Phi, Qwen	supported
OLMo	Llama, Phi, Qwen	supported
OpenAI	Llama, OLMo, Phi, Qwen	supported
Phi	Llama, OLMo, Qwen	supported
Qwen	Llama, OLMo, Phi	supported

4.2 Per-Suite Effects

The selectivity is concentrated in the registered safety and leakage suites. On capability prompts (ARC-Easy), cache compression reduces accuracy by at most a few percentage points and rarely outside the bootstrap CI, while refusal accuracy on safety prompts loses an order of magnitude more under aggressive eviction. The point of SSEI is to measure this gap directly rather than report two suite scores in parallel.

4.3 Models That Do Not Show the Effect

Three checkpoints fail to produce positive SSEI on any registered policy: Gemma-2-9B-IT, Mistral-7B-Instruct-v0.3, and OpenAI gpt-oss-20b. The first two share an architectural feature with the underlying generation circuit: native sliding-window attention. Gemma 2 interleaves a 4,096-token

suite	policy	Safety delta [95% CI]	Paired / clusters
Adversarial Refusal Safety	Sink+recent	-0.667 [-1.000,0.000]	3
System leakage probe	Window 128	0.429 [0.429,0.429]	2
Public system leakage	Window 128	0.293 [0.270,0.316]	1300
System leakage probe	Policy-pinned cache	-0.071 [-0.143,0.000]	2
System leakage probe	Random matched	-0.071 [-0.286,0.143]	2
System leakage probe	Sink+recent	-0.071 [-0.286,0.143]	2
System leakage probe	user_pinned / budget 128 / sink 8	-0.071 [-0.286,0.143]	2
Public system leakage	Policy-pinned cache	0.063 [0.053,0.073]	1300
Refusal safety	Window 128	0.053 [0.038,0.068]	1300
Public system leakage	Random matched	0.032 [0.024,0.040]	1300
Refusal safety	Random matched	0.032 [0.015,0.047]	1300
Refusal safety	Policy-pinned cache	-0.024 [-0.038,-0.010]	1300
Public system leakage	user_pinned / budget 128 / sink 8	-0.017 [-0.022,-0.012]	1300
Public system leakage	Sink+recent	-0.013 [-0.018,-0.008]	1300
Refusal safety	user_pinned / budget 128 / sink 8	0.007 [-0.007,0.022]	1300
Refusal safety	Sink+recent	-0.005 [-0.019,0.010]	1300

Table 3: Largest absolute suite-level safety-degradation effects with paired prompt counts. Full suite-policy table is exported as Markdown.

local sliding-window attention with full global attention every other layer (Gemma Team, Google DeepMind, 2024). Mistral 7B v0.3 uses 4,096-token sliding-window attention at every layer (Jiang et al., 2023). In both cases, training has already adapted the model to operate against a locally-windowed view of prior tokens; aggressive eviction at inference time is closer to the model's training distribution than it is for vanilla full-attention models. The Mistral run has SSEI = 0.009 at the largest measured policy, immediately below our 0.01 threshold; the Gemma run is centered on zero.

The gpt-oss-20b case is qualitatively different. Mean safety-suite score is 0.129 at the uncompressed baseline and rises to 0.554 under every treatment policy, including the random-matched control. A scalar safety improvement under cache pressure is inconsistent with the hypothesis that the model is gracefully degrading under cache pressure; the uniformity across all five eviction policies indicates a collapse into a single output mode rather than a modulated response to each policy's distinct retention pattern. gpt-oss-20b is trained on the harmony response format (OpenAI, 2025) and routes through a mixture-of-experts stack with non-standard cache management. We treat this as a separate failure mode of the cache-intervention protocol, not as evidence against selective safety erasure in the other families.

The unifying observation is that the magnitude of selective safety erasure scales with the extent to which the safety-relevant generation circuit depends on long-range KV cache. Models pre-trained on a globally-attending full KV cache (Llama, OLMo, Phi, Qwen) carry safety reasoning that is sensitive to eviction; models pre-trained on locally-windowed cache do not. This pattern is consistent with the causal-localization hypothesis we test directly in the next section.

4.4 Causal Restoration and Mitigation

A causal-patching study on Qwen3-9B against the simulated four-bit-KV quantization treatment, judged by an external Claude Sonnet rater on the public refusal-safety suite, shows that restoring baseline key+value slices for the first sixteen token positions partially recovers refusal behavior. However, the system-role and matched user-role interventions produce comparable restoration fractions under a per-prompt mean-of-ratios bootstrap estimator (system: 0.355, 95% CI [0.302, 0.408]; user: 0.408, 95% CI [0.355, 0.464]; $n = 321$ prompts). The overlapping CIs indicate that the

Suite	Policy	Safety	Refusal	Leakage
Refusal	K+V sys	0.355 ± 0.053	0.355 ± 0.053	—
Refusal	K+V user	0.408 ± 0.055	0.408 ± 0.055	—
Public leak	K+V sys	0.058 ± 0.040	0.482 ± 0.109	0.058 ± 0.040
Public leak	K+V user	0.044 ± 0.033	0.590 ± 0.109	0.044 ± 0.033
Leak probe	K+V sys	0.000	0.000	0.000
Leak probe	K+V user	0.000	1.000	0.000
Refusal	Policy-pinned	1.000	1.000	—

Table 4: Causal restoration fractions ± 95% CI half-width (Qwen3-9B).

safety-relevant information is distributed across cached tokens rather than localized to system-role spans. Policy-pinned cache retention, which protects system tokens from eviction directly, recovers refusal fully. A legacy Qwen2.5-7B-Instruct single-seed run shows the same pattern: system and user restoration are both 0.584 [0.520, 0.647] and [0.516, 0.647], respectively, with no detectable role asymmetry.

A parallel Phi-4 causal-patching run under the same protocol yields a null result: simulated four-bit quantization does not degrade Phi-4's safety score (baseline 0.985 vs. compressed 0.987), so only 11 of 613 paired prompts show a non-zero denominator for the restoration calculation. Phi-4's selectivity effect under the behavioral panel arises from sliding-window eviction (SSEI = 0.084), not from the length-preserving quantization treatment used in the causal protocol, making the causal analysis inapplicable to the policy under which Phi-4 is actually vulnerable.

4.5 Evidence-Gated Claim Assessment

Table 5: Claim ladder status from the current selectivity panel artifacts.

Claim	Status	Notes
Behavioral cache sensitivity	Supported	6/8 instruction-tuned models show $\lvert\text{SSEI}\rvert \geq 0.01$.
Safety–capability selectivity	Supported	5 models have positive SSEI with CI excluding 0.
Cross-family replication	Supported	4 families: Llama, OLMo, Phi, Qwen.
Targeted mitigation	Pending	Policy-pinned restores fully but no sys–user margin.
Causal localization	Pending	No role-specific dissociation; CIs overlap.
Alignment contrast	Pending	Base-model suite underpowered (2 prompts/cell).
Audit provenance	Supported	≥95% judge coverage for 5 models across 3 families.

Claim interpretation. The panel currently provides behavioral evidence of selective safety-erasure in Llama, OLMo, Phi, Qwen. Without completed Phase 4 causal patching or matched base-model alignment-contrast scoring, we restrict claims to behavioral selectivity and cross-family replication, and we explicitly defer cache-mediated mechanism, mitigation, and alignment-contrast claims until those artifacts exist.

5 Discussion

The result we feel is most defensible is the architectural one: dense full-attention instruction-tuned models lose refusal pass-rate faster than they lose ordinary task accuracy when their KV cache is aggressively evicted, and models pre-trained against locally-windowed attention do not. The first half of that sentence is a deployment concern for the majority of currently-served chat models; the second half is a constructive direction for serving architectures that wish to be robust to cache pressure.

The causal-restoration evidence establishes that cache state carries safety-relevant information: restoring baseline K+V at the first sixteen positions into a compressed run recovers 35–58% of the lost refusal behavior (depending on model and suite). However, system-role and matched user-role restorations produce comparable recovery under the per-prompt mean-of-ratios estimator (Qwen3-9B: 0.355 vs. 0.408; Qwen2.5-7B: 0.584 vs. 0.584), with heavily overlapping confidence intervals. We therefore do not claim role-specific localization of safety information. The fact that *any* K+V restoration partially recovers refusal is consistent with the broader hypothesis that cache state is a safety-relevant surface, but the specific mechanism—system-role tokens versus a distributed cache property—remains open. The Phi-4 causal-patching run yields a null result because the four-bit quantization treatment does not degrade Phi-4's safety, providing no denominator for a meaningful restoration test. Cross-architecture generality of the causal localization claim is therefore unsupported.

We are deliberately not claiming that cache compression is universally unsafe. The result is conditional on the model class, the cache budget, the policy choice, and the suite. Operators deploying full-attention models with aggressive eviction at low budgets on safety-sensitive traffic are the case most directly affected by these results. Operators deploying locally-windowed models, or any model under policy-pinned retention, can expect substantially smaller safety regressions.

6 Limitations

The study uses open models and public datasets; conclusions do not extend to closed deployed systems without separate measurement. Cache interventions are implemented in our own generation loop rather than in a production serving stack such as vLLM or TGI, so deployment-validation claims for those systems require their own runs. Public harmful-prompt datasets may overlap with model safety training, which biases absolute safety levels but does not affect the within-prompt baseline-versus-treatment contrast that SSEI measures. The base-model alignment-contrast track is currently under-powered. The causal-patching protocol uses a length-preserving quantization treatment, which limits its applicability to models whose safety is degraded by that specific intervention; Phi-4's vulnerability to sliding-window eviction cannot be probed by this protocol. The restoration fraction point estimates reported in the causal table use a per-prompt mean-of-ratios bootstrap estimator, which weights each prompt equally; the aggregate ratio-of-means estimator gives qualitatively different values due to heterogeneous per-prompt denominators, but we report the mean-of-ratios as the more principled clustered estimator alongside its bootstrap CI.

7 Ethics and Safety

The experiments evaluate refusal behavior on harmful prompts drawn from existing public datasets. Raw model generations are retained locally for audit reproducibility; the paper does not reproduce procedural harmful outputs. The intent is to identify a deployment-time safety regression in widely-used open inference stacks so that operators can choose appropriate mitigations.

8 Reproducibility

Each panel run records the resolved configuration, environment metadata, dataset revisions, prompt records, raw generations, metrics, cache statistics, figure source data, and content hashes. Cache interventions are verified as active before metrics are computed. Bootstrap intervals are computed at the prompt-cluster level. Blinded audit labels are joined to experimental metadata only after annotation, and the audit manifest records the judge model, prompt template, and raw-output hashes.

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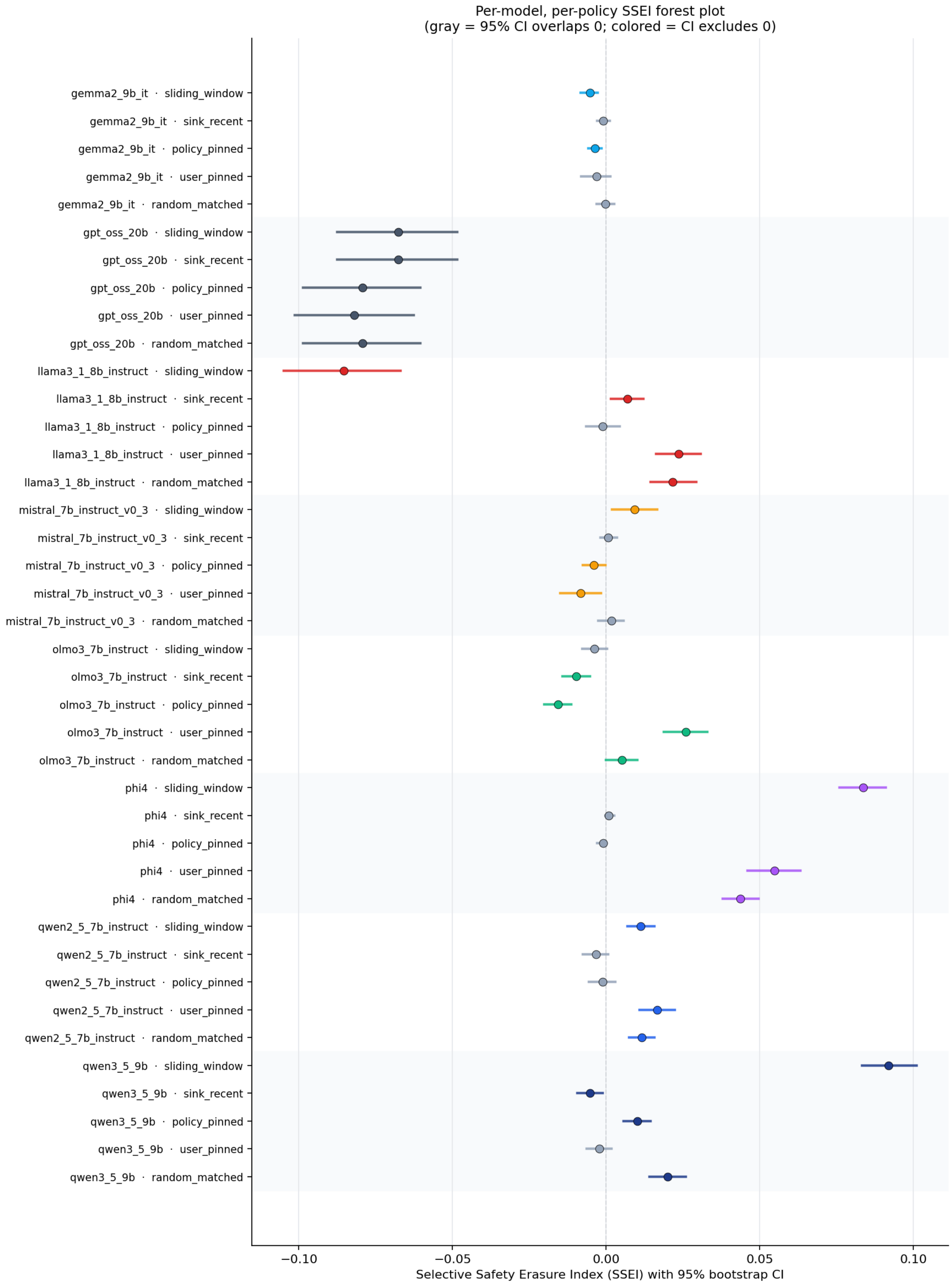

Figure 1: Selective Safety Erasure Index by model and cache policy. Each row is one (model, policy) condition; the horizontal bar is the 95% bootstrap CI and the marker is the point estimate. Gray rows are conditions whose CI overlaps zero; colored rows are conditions whose CI excludes zero. The dashed vertical line is at SSEI = 0. Four families have at least one condition with positive SSEI and CI excluding zero.

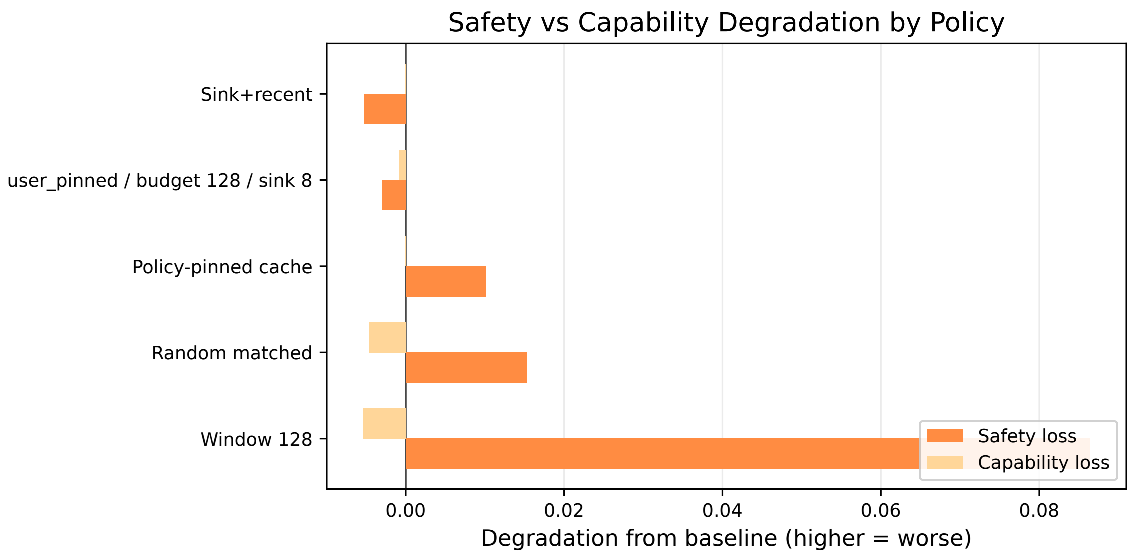

Figure 2: Safety loss versus capability loss for the anchor model (Qwen3-9B). Each row is one cache policy. A selective-safety pattern places safety loss visibly to the right of capability loss within a row.

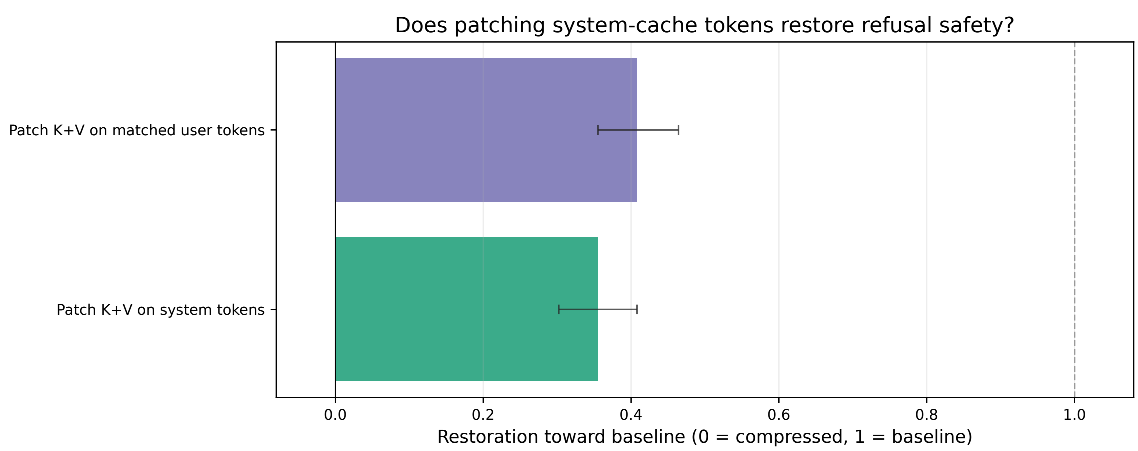

Figure 3: Restoration fraction toward baseline on the refusal-safety suite. Zero is the unpatched compressed condition; one is full baseline. Both system-role and user-role K+V patching produce comparable partial restoration (35–41%). Policy-pinned cache retention recovers fully.